Sushanth Buddhala

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PROFILE

Data Analytics Engineering graduate with hands-on experience building agentic, LLM-powered systems and retrieval pipelines, skilled in Python, SQL, and prompt engineering. Experienced designing multi-agent orchestration, evaluation harnesses, and data pipelines, with a foundation in networking and infrastructure automation and a focus on turning complex technical work into reliable, production-ready systems.

WORK EXPERIENCE

Data Analyst - QA & Security - Community Dreams Foundation *Oct 2026 - Present*

- Test CitadelTwin, an AI platform for construction site assessment and 2D-to-3D blueprint conversion, end to end with test cases and regression checks before releases.
- Validate data and model outputs across site, regulatory, and 3D conversion workflows, and log reproducible bugs for the development and ML teams.
- Review the application and its APIs for vulnerabilities, including authentication, access control, and input handling.

Software Engineering Intern - Apposite Technologies LLC *Jul 2026 - Sep 2026*

- Built a Python CLI that provisions a 3-VM Netropy VE network-emulation lab across 3 subnets from a single JSON file, using OpenTofu, KVM/libvirt and cloud-init. A schema and validation layer catches bad IPs, overlapping subnets and sizing errors before anything is created, backed by 40+ pytest unit tests.
- Automated lab health checks and Ansible-based route setup, then confirmed end-to-end traffic through the emulator with iperf3 at ∼7 Gbit/s TCP, 0% ping loss and ∼0.6 ms average latency; the full lab tears down in ∼11 seconds.
- Diagnosed and fixed 10 deployment blockers across KVM and VMware ESXi (kernel panics, missing data ports, disk-permission failures, vSwitch security policies), rebuilt the lab by hand on ESXi, and wrote the tool comparison, runbook and handover docs.

PROJECTS

Compass - Agentic Multi-Agent Retail Analytics System *Feb 2026 – May 2026*

- Architected a 5-agent system (Planner, Data, Product, Supply Chain, Synthesizer) in Python using Gemini API function calling and Pydantic orchestrator on DuckDB, built with Cursor AI assistance.
- Built a column synonym normalization layer, hybrid Flash/Pro fallback strategy, and hand-crafted system prompts for all 5 agents through iterative prompt engineering; deployed a Streamlit chat UI with inline Plotly charts and reasoning trace visualization.
- Designed a 40-case integration eval harness with 4-dimensional scoring (correctness, routing, SQL quality, overall) using Pytest, reaching 95% completion and 95.2% correctness across 3 iterative bug-fix rounds.

SARRDI- Dam Infrastructure Risk Analytics (Capestone Project) *Jan 2026 – May 2026*

- Engineered an ETL pipeline for Virginia DCR, integrating inspection, flood, and social vulnerability data across 2,712 dams via fuzzy matching and watershed-level joins into a unified risk dataset.
- Built a 5-dimension weighted risk-scoring model and Power BI/DAX dashboard under GMU's STAR-TIDES Program, flagging 170 dams as critical priority.
- Delivered field-ready decision support to DCR with zero change requests, supporting statewide dam-safety prioritization.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, DAX, Pandas, NumPy, Scikit-learn, Pydantic, FastAPI, Streamlit, LangChain, Pytest
Data & BI Platforms: Snowflake, Databricks, DuckDB, SQLite, AWS, Power BI, Tableau, Microsoft Excel, Google Sheets, Git, Linux
Data Engineering & Analytics: Data Modeling, Data Warehousing, ETL Pipelines, Data Validation, Data Analysis, Statistical Analysis, KPI Reporting
AI & LLM: Gemini API, OpenAI API, Function Calling, Prompt Engineering, Multi-Agent Systems, Workflow Orchestration, Model Evaluation, RAG Pipelines, Generative AI, Cursor AI
Testing & Security: Quality Assurance, Manual Testing, Application Security, API Security

EDUCATION

George Mason University Aug 2024 – May 2026
Master of Science in Data Analytics Engineering

Gokaraju Rangaraju Institute of Engineering and Technology Nov 2020 – Jul 2024
Bachelor of Science in Computer Science Engineering